

MOLODYOZHNYA STATION, Antarctica

WMO: 895420

Lat: 67.666S

Lon: 45.850E

Elev: 50

StdP: 100.73

Time Zone: 3.00 (E03)

Period: 82-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	-30.1	-28.0	-34.6	0.1	-28.8	-32.9	0.2	-27.8	N/A	N/A	N/A	N/A	N/A	140	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.1	3.6	0.2	2.5	-0.5	1.6	-1.2	0.5	3.0	-0.2	1.9	-0.8	1.2	N/A	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
-1.5	3.4	0.1	-2.3	3.1	-0.1	-3.0	3.0	-0.4	10.7	3.3	9.4	2.1	8.4	1.3	4.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			-32.8	1.7	3.4	1.1	-35.3	2.5	-37.3	3.1	-39.2	3.7	-41.6	4.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-11.1	-1.1	-3.7	-8.2	-11.8	-15.3	-15.5	-16.7	-18.4	-18.3	-13.9	-6.9	-2.2		
	DBStd	7.38	2.01	2.70	3.76	4.40	4.78	4.79	5.29	5.42	5.24	4.27	3.50	2.19		
	HDD10.0	7684	343	384	565	656	786	765	828	880	849	742	507	379		
	HDD18.3	10726	601	618	823	906	1044	1015	1087	1138	1099	1000	757	638		
	CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(15) Precipitation	PrecAvg	454	34	40	41	41	38	43	44	38	30	39	37	30		
	PrecMax	687	79	94	69	70	81	103	98	73	63	113	82	74		
	PrecMin	284	7	21	10	10	16	11	14	11	7	12	11	6		
	PrecStd	102	17	19	12	14	15	22	21	18	14	23	20	15		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.1	3.2	-0.3	-1.6	-2.5	-4.3	-4.0	-5.3	-6.0	-3.5	1.9	4.4		
		MCWB	1.2	0.2	-1.9	-2.9	-3.2	-5.5	-4.9	-6.7	-7.2	-4.3	-1.6	0.6		
	2%	DB	4.1	1.6	-1.4	-3.2	-5.3	-5.4	-5.6	-7.0	-7.1	-5.3	-0.1	3.1		
		MCWB	0.5	-1.2	-2.5	-4.2	-6.1	-6.5	-6.5	-7.9	-7.9	-6.5	-2.8	-0.2		
	5%	DB	3.1	0.6	-2.3	-4.4	-6.5	-6.6	-7.0	-8.8	-8.8	-6.6	-1.3	2.1		
		MCWB	-0.1	-1.7	-3.3	-5.3	-7.1	-7.6	-7.7	-9.6	-9.5	-7.6	-3.6	-0.9		
	10%	DB	2.1	-0.2	-3.2	-5.5	-8.1	-8.1	-8.9	-10.7	-10.5	-8.0	-2.3	1.2		
		MCWB	-0.8	-2.1	-4.2	-6.5	-9.0	-9.1	-9.8	-11.5	-11.3	-8.9	-4.2	-1.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.7	0.4	-1.3	-2.4	-3.5	-5.3	-4.7	-6.5	-6.9	-4.1	-1.4	0.9		
		MCDB	4.7	2.4	-1.0	-1.8	-3.0	-5.0	-4.4	-5.5	-6.4	-3.6	0.4	4.0		
	2%	WB	0.9	-0.5	-2.2	-3.9	-5.8	-6.5	-6.2	-7.8	-7.8	-6.2	-2.5	0.0		
		MCDB	3.6	0.8	-1.4	-3.2	-5.5	-5.7	-5.4	-7.0	-7.1	-5.5	-0.8	2.4		
	5%	WB	0.2	-1.3	-3.0	-5.1	-7.0	-7.6	-7.5	-10.0	-9.4	-7.4	-3.5	-0.6		
		MCDB	2.6	0.1	-2.2	-4.4	-6.4	-6.7	-6.9	-9.3	-8.7	-6.6	-1.8	1.6		
	10%	WB	-0.4	-2.0	-4.0	-6.5	-8.8	-9.2	-9.5	-12.3	-11.2	-8.8	-4.3	-1.3		
		MCDB	1.8	-0.4	-3.2	-5.6	-8.1	-8.2	-8.7	-11.6	-10.6	-8.0	-2.7	0.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.1	4.3	4.6	4.7	4.2	4.6	4.5	5.2	5.2	5.4	5.1	4.1		
		MCDBR	5.1	4.3	3.9	5.1	5.1	5.1	5.0	5.7	6.4	5.6	6.4	4.9		
	5% WB	MCWBR	3.6	3.4	3.9	4.6	5.0	5.0	5.0	5.7	6.4	5.2	5.1	3.9		
		MCWBR	4.6	3.7	3.6	4.5	5.1	5.2	4.8	6.3	6.5	5.0	5.6	4.5		
(40) Clear-Sky Solar Irradiance	taub		0.255	0.249	0.226	0.186	0.125	N/A	0.125	0.170	0.200	0.218	0.236	0.255		
		taud	2.234	2.282	2.303	2.164	1.901	N/A	1.879	2.107	2.216	2.218	2.228	2.205		
	Ebn at Noon		1000	947	868	669	241	0	202	679	889	980	1019	1015		
		Edh at Noon	117	97	71	40	9	0	8	40	74	101	117	126		
(44) All-Sky Solar Radiation	RadAvg		7.76	4.99	2.58	0.78	0.10	0.00	0.03	0.47	2.03	4.52	7.37	8.88		
	RadStd		0.34	0.32	0.22	0.06	0.01	0.00	0.00	0.03	0.10	0.16	0.21	0.27		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page